Kumar Kartikeya

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Software Engineer At Capgemini | **LeetCode Guardian 2404** kumarkartikeya877@gmail.com
Codeforces Expert 1700+ | Peak Rank **300/40K** | **Top 0.37% Globally** linkedin.com/in/kumar-kartikeya
CodeChef 5 2077 | Peak Rank **35/20K**

Summary

Full Stack **Java Backend Developer** with **2.1 years** of experience delivering **scalable**, **high-availability** financial applications across **Hong Kong** and **Singapore** markets using **Java** and **Spring Boot**. Proven expertise in **RESTful API development**, **Microservices** architecture, **distributed systems**, and **performance optimization** using **Redis** and **SQL**. Experienced in **CI/CD pipelines**, **cloud-native** deployments on **AWS**, and **Agile** delivery in cross-functional banking teams. Actively integrates **Generative AI** tooling to accelerate production-ready backend solutions.

Technical Skills

Core Java: OOP, Collections Framework, Multithreading, Concurrency, Design Patterns, Data Structures & Algorithms, Java 8
System Design: High-Level Design (HLD), Low-Level Design (LLD), Distributed Systems, Scalable Architecture
Backend: Spring Boot, Spring MVC, Spring Data JPA, Spring Security, Hibernate, REST APIs, Microservices
Messaging & Cache: Apache Kafka, Redis, Event-Driven Architecture
Database: MySQL, PostgreSQL, JPA, Hibernate, SQL Query Optimization
Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD Pipelines, Git, Maven
Security & Testing: JWT, OAuth 2.0, Spring Security, JUnit 5, Mockito
AI / GenAI: GitHub Copilot, Claude API, OpenAI API, Prompt Engineering
Frontend: React, Angular, TypeScript, JavaScript

Experience

Capgemini Bengaluru, India
Software Engineer

- **Enhanced Mobile Wallet Integration Performance** Optimized JETCO registration & mobile wallet processing pipelines, reducing API response time by 20% and improving transaction throughput for 50,000+ active HK users.
- **Streamlined Account Opening Workflow** Engineered RESTful APIs for the IAO (Investment Account Opening) module, achieving 25% faster processing with 99.2% uptime across Hong Kong and Singapore markets.
- **Modernized Build and Deployment Pipeline** Refactored Maven build system and CI/CD configurations, reducing bundle size by 35% and cutting deployment time from 45 to 12 minutes, enabling faster release cycles.
- **Bilingual Chatbot Feature Engineering** Engineered context-aware iBanking chatbot with English/Chinese bilingual support using JavaScript, reducing average user navigation time by 35% through instant, context-specific menu rendering for 10,000+ monthly active users.
- Conducted 7+ technical knowledge-sharing sessions and peer code reviews, increasing clean code compliance and reducing post-deployment defects by 15% across the iBanking module.

Projects

Production-Ready API Rate Limiter (Java, Spring Boot, Redis, Docker, JUnit 5) **GitHub**

- Leveraged **Java Multithreading** — **synchronized** blocks, **AtomicLong**, and **ReentrantLock** — across 20 concurrent threads, achieving zero data corruption under high-concurrency load tests.
- Applied **Collections Framework** (**ConcurrentHashMap**, **ConcurrentLinkedDeque**) for thread-safe, per-identifier request tracking with zero race conditions at peak load.
- Implemented 4 algorithms using core **Data Structures** — **Deque** (Sliding Window Log), **LinkedHashMap** (Fixed Window), **ConcurrentHashMap** (Token Bucket) — achieving **O(1)** average-case evaluation.
- Built distributed rate limiting via atomic **Redis Lua scripts** with TTL-based expiration and in-memory fallback, ensuring **99.9% uptime** across microservice instances.
- Delivered **100% test pass rate** (13 unit + 7 integration) via **JUnit 5**; enabled hot-swap across all 4 algorithms using **Strategy Pattern** with DB-override + YAML config.

Education

Haldia Institute of Technology **Haldia, West Bengal**
B.Tech in Computer Science and Engineering — 8.91 DGPA July 2019 – June 2023
Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Distributed Systems

Achievements

Recognized by the client as the **Best YP in the Team** for consistently delivering high-quality work and meeting project expectations.
Completed Industrial Training Program on **Java programming, Data Structure and Algorithms and, Database Management System** from **MYWBUT.**
Solved **200+ Java** and **100+ SQL** problems on **LeetCode, GeeksforGeeks, and HackerRank**, strengthening problem-solving skills and data structures & algorithms, covering joins, subqueries, window functions, CTEs, and query optimization.